AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

OZONATION SYSTEM

EFFECTIVE DISINFECTION & ADVANCED OXIDATION

FOR WATER & WASTEWATER TREATMENT

- Stronger Disinfection
- Removal of Odour & Colour
- Oxidation of Organics
- Improved Water Reuse
- Chemical Free & Eco-friendly
- Reliable & Low Maintenance



Ozone Generator Panel (1 – 500 gm/hr)



Ozone Contact Tank SS-304 / FRP / MSEP



Oxygen Generation System (PSA)



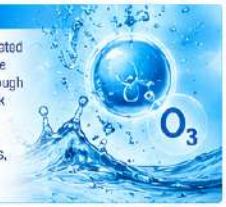
Ozone Mixing System (Venturi / Diffuser)

PROCESS FLOW DIAGRAM – OZONATION SYSTEM



WORKING PRINCIPLE

Ozone (O_3) is a powerful oxidizing agent, generated from high purity oxygen using corona discharge technology. The ozone is mixed with water through venturi/diffuser and contacted in a contact tank where it inactivates bacteria, virus and other microorganisms. It also oxidizes colour, odour, organics, iron, manganese and other impurities, resulting in high quality treated water.



KEY FEATURES

- Ozone generation from pure oxygen (PSA / Cylinder)
- Automatic ozone dose control
- High disinfection efficiency (bacteria, virus, pathogen)
- Removal of odour, colour and taste
- Oxidation of BOD/COD, iron, manganese and sulphide
- PLC / HMI based automation (optional)
- Low operating cost and easy maintenance
- Compact and modular design
- Suitable for continuous or intermittent operation
- Available in SS-304 / FRP / MSEP

APPLICATIONS / INDUSTRIES



Residential Societies



Hotels & Resorts



Hospitals & Healthcare



Industrial Effluents



Food & Beverage Industry



Pharmaceutical



Dye & Chemical Industry



STP Tertiary Treatment



Water Reuse Projects



Swimming Pools



Municipal & Sewage Treatment

TECHNICAL SPECIFICATIONS – OZONATION SYSTEM (1 gm/hr to 500 gm/hr)

Model	Ozone Capacity (gm/hr)	Oxygen Requirement (Nm^3/hr)	Ozone Concentration (wt %)	Ozone Contact Tank Volume (min)	Power Load (kW)	Suggested Flow Range (m^3/hr)*
AQU-OZ-1	1	0.1	6 – 8	0.25 – 0.5	0.5	0.5 – 2
AQU-OZ-5	5	0.5	6 – 8	0.5 – 1.0	1.0	1 – 5
AQU-OZ-10	10	1.0	6 – 8	1 – 2	1.5	2 – 10
AQU-OZ-20	20	2.0	6 – 8	2 – 4	3.0	5 – 20
AQU-OZ-50	50	5.0	6 – 8	5 – 10	6.0	10 – 50
AQU-OZ-100	100	10.0	6 – 8	10 – 20	12.0	20 – 100
AQU-OZ-200	200	20.0	6 – 8	20 – 40	22.0	50 – 200
AQU-OZ-300	300	30.0	6 – 8	30 – 60	32.0	100 – 300
AQU-OZ-500	500	50.0	6 – 8	50 – 100	50.0	200 – 500

Note: Flow range is indicative for 0.1 – 0.5 mg/L residual ozone depending on application and contact time.

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet	Typical Outlet (after Ozonation)
pH	-	6.0 – 8.5	6.0 – 8.5
BOD	mg/L	10 – 100	< 10
COD	mg/L	50 – 300	< 30
TSS	mg/L	10 – 100	< 10
Colour (Pt-Co)	mg/L	50 – 500	< 10
Odour	-	Present	Removed
Total Coliform	MPN/100 mL	$10^6 – 10^8$	< 10^3
E. Coli	MPN/100 mL	$10^4 – 10^6$	< 10^2
Iron (Fe)	mg/L	0.5 – 5.0	< 0.1
Manganese (Mn)	mg/L	0.1 – 2.0	< 0.1
Ammonia (as N)	mg/L	5 – 50	< 5

Note: Outlet values can be customized as per application and dosage.

MATERIAL OF CONSTRUCTION (MOC)

Item	MOC
Ozone Generator Enclosure	SS-304 / MS with Powder Coating
Ozone Reactor	SS-304 / Ti (Optional)
Contact Tank	SS-304 / FRP / MSEP with FRP Lining
Piping (Ozone Line)	SS-316 / UPVC / PVDF (as per application)
Piping (Water Line)	UPVC / CPVC / SS-304
Venturi / Diffuser	UPVC / SS-304 / PVDF
Non Return Valve	SS-316 / UPVC / CPVC
Isolation / Control Valves	UPVC / CPVC / SS-304
Flow Meter / ORP / Ozone Sensor	PP / PVDF / SS-304
Control Panel	MCC with PLC & HMI (Optional)
Oxygen Generation System (PSA)	MS / SS-304

TYPICAL BILL OF MATERIAL (BOM) – OZONATION SYSTEM



Oxygen PSA System



Ozone Generator with Panel



Ozone Contact Tank



Venturi/Diffuser Mixing System



Booster Pump



Flow Meter



ORP/Ozone Sensor



Instrumentation



Piping, Valves & Fittings



Control Panel (PLC/HMI)

OPTIONAL ACCESSORIES



PSF



ACF



Micron Filter



Online Ozone Residual Analyzer



Ozone Destroyer (Catalytic)





UV (as backup)

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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SUSTAINABLE WATER SOLUTIONS FOR A BETTER TOMORROW

1. CIVIL WORK IN PROGRESS – STP + TERTIARY TREATMENT (PSF, ACF, OZONE) – 100 KLD



2. STP WITH OZONE DISINFECTION – 100 KLD MSRL PACKAGE PLANT – INSTALLATION IN PROGRESS



3. PRE TREATMENT – MGF, ACF, SOFTENER WITH OZONE FOR RECYCLED WATER – 50 KLD



4. ELECTRO-MECHANICAL INSTALLATION – STP + FILTERS + OZONE 200 KLD



5. OZONATION SYSTEM – INSTALLATION IN PROGRESS (5 – 50 g/hr, SKID MOUNTED)



6. COMPLETE PLANT – STP + MGF + ACF + SOFTENER + OZONE (100 KLD) – SITE IN PROGRESS

